

Medical Card Certificate

wrought havoc in China and caused a pandemic situation in the worldwide population, leading to disease outbreaks that have not been controlled to date, although extensive efforts are being put in place to counter this virus (25). This virus has been proposed to be designated/named severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by the International Committee on Taxonomy of Viruses (ICTV), which determined the virus belongs to the *Severe acute respiratory syndrome-related coronavirus* category and found this virus is related to SARS-CoVs (26). SARS-CoV-2 is a member of the order *Nidovirales*, family *Coronaviridae*, subfamily *Orthocoronavirinae*, which is subdivided into four genera, viz., *Alphacoronavirus*, *Betacoronavirus*, *Gammacoronavirus*, and *Deltacoronavirus* (3, 27). The genera *Alphacoronavirus* and *Betacoronavirus* originate from bats, while *Gammacoronavirus* and *Deltacoronavirus* have evolved from bird and swine gene pools (24, 28, 29, 275).

Coronaviruses possess an unsegmented, single-stranded, positive-sense RNA genome of around 30 kb, enclosed by a 5'-cap and 3'-poly(A) tail (30). The genome of SARS-CoV-2 is 29,891 bp long, with a G+C content of 38% (31). These viruses are encircled with an envelope containing viral

Driver Acknowledgement



Signature

2026-04-30  
Signature Date

